Analysis of Symmetry of 2-point Correlation Functions in Finite Temperature QCD

Using Möbius Domain-Wall Fermions

Junxiong Nie, U. Osaka
for JLQCD Collaboration

Outline

- Research Background & Motivation
- Theoretical Background
- Lattice QCD Simulation Setup
- Chiral Condensate & Susceptibility Analysis
- Meson Screening Masses & Symmetry Restoration
- Summary and Outlook

Research Background

- According to **Pisarski & Wilczek (1984)**, the restoration of $U(1)_A$ axial symmetry around the chiral phase transition is crucial for determining the order and universality class of the transition.
- Previous lattice studies often relied on staggered or Wilson-type fermions that explicitly break chiral or taste symmetries at finite lattice spacing a.
- Our recent $N_f = 2$ study by the **JLQCD Collaboration** [Phys. Rev. D **111**, 114506 (2025)] using Möbius Domain-Wall Fermions (MDWF) showed that both $SU(2)_L \times SU(2)_R$ chiral symmetry and $U(1)_A$ symmetry are effectively restored near $T_c \approx 165(3)$ MeV.

Motivation & Scope of This Work

- **Realistic $N_f = 2 + 1$ QCD with MDWF**: Extend our investigation by including the dynamical strange quark while preserving lattice chiral symmetry ($L_s = 12$).
- **Physical Point Temperature Scan ($48^3 \times 16$)**:
  - Measure the renormalized subtracted chiral condensate $\Delta^R_{l,s}(T)$ and disconnected chiral susceptibility $\chi_{\text{disc}}(T)$ across 8 couplings $\beta \in [4.13, 4.405]$ ($T \in [138.2, 241.6]$ MeV) to pin down T_{pc}.
  - Extract spatial meson screening masses across 6 channels using 16-source and directional averaging.
- **Fixed $\beta = 4.17$ Multi-Mass & Finite-N_t Scan**:
  - Investigate quark mass dependence ($am_l \in \{0.0020, 0.0035, 0.0070, 0.0120\}$, $am_s = 0.040$) across $N_t \in \{12, 14, 16, 18\}$ ($T \in [136.0, 204.0]$ MeV, $a^{-1} \approx 2453$ MeV) on lattices up to $48^3 \times 18$ and $40^3 \times 16$.

Symmetry of QCD

QCD Partition Function: $\mathcal{Z} = \int \mathcal{D}A\mathcal{D}\bar{q}\mathcal{D}q e^{-S}$, Gauge Boson + Fermion

$$S = \int d^4x \left[-\tfrac{1}{2g^2} \mathrm{Tr} F^{\mu\nu} F_{\mu\nu} + \bar{q}_a (i\not{D} - m_a) q_a \right], \quad a \in [1, N_f]$$

In the massless up-down quark limit ($m_{u,d} \to 0$):
Flavored chiral symmetry $SU(2)_L \times SU(2)_R$:

$$q \to \exp(i\alpha_a T^a + i\beta_a T^a \gamma_5) q, \quad \bar{q} \to \bar{q} \exp(-i\alpha_a T^a + i\beta_a T^a \gamma_5),$$

$U(1)_A$ Axial Symmetry (broken at the quantum level by the ABJ anomaly):

$$q \to \exp(i\epsilon\gamma_5) q, \quad \bar{q} \to \bar{q} \exp(i\epsilon\gamma_5).$$

Possible Emergent Larger Symmetry

- Dimensional reduction / NRQCD_3 perturbative analysis [Laine & Vepsäläinen (2004)] predicts a **spin-independent** asymptotic screening mass at high T due to the large Matsubara mass $M \approx \pi T$:

$$m_{\mathrm{scr}} = 2\pi T + g_E^2 \frac{C_F}{2\pi}\left(\frac{1}{2} + \hat{E}_0\right) \approx 2\pi T + 0.1872 g^2 T$$

- $U(1)_A$ is expected to be restored at high temperature as instanton/topological fluctuations are suppressed.
- Furthermore, an emergent larger symmetry, **Chiral-Spin Symmetry** $SU(2)_{CS}$, has been proposed in the regime where chromoelectric interactions dominate over chromomagnetic ones:

$$q \to \exp(i\Sigma^i \epsilon_i) q,$$
$$\bar{q} \to \bar{q}\gamma_0 \exp(-i\Sigma^i \epsilon_i)\gamma_0, \qquad \text{where } \Sigma^i = (\gamma_k, -i\gamma_5\gamma_k, \gamma_5), \quad k = 1, 2.$$

Meson Correlation Functions

We consider the following set of flavor non-singlet meson operators:

$$O_\Gamma(x) = \bar{q}(x)\Gamma F^a q(x),$$

$$\Gamma \in \left\{ \mathbf{1}(S),\ \gamma_5(PS),\ \gamma_\mu(V),\ \gamma_\mu\gamma_5(A),\ \sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu, \gamma_\nu](T),\ \sigma_{\mu\nu}\gamma_5(X) \right\}$$

We measure the spatial two-point correlation functions along spatial directions (x, y, z). By periodicity on a lattice of spatial extent $L = N_s a$, the asymptotic behavior at large spatial separation is governed by the screening mass m_Γ:

$$C_\Gamma(x) \simeq A \cosh\left(m_\Gamma \left(x - \frac{N_s}{2} \right) \right)$$

Symmetry of Correlation Functions

- Flavored chiral symmetry $SU(2)_L \times SU(2)_R$: $\delta_C O^a_\Gamma = \frac{i}{2}\beta_b \bar{q}\left(\{\Gamma,\gamma_5\}\frac{1}{2}\delta^{ab}\mathbf{1} + [\Gamma,\gamma_5] i\epsilon^{bac}\frac{\tau^c}{2}\right) q$
- $U(1)_A$ Symmetry : $\delta_A O^a_\Gamma = i\epsilon\bar{q}\{\Gamma,\gamma_5\}\frac{\tau^a}{2} q$

Γ	$[\Gamma,\gamma_5]$	$\{\Gamma,\gamma_5\}$
$\mathbf{1}(S)$		γ_5
$\gamma_5(P)$		$\mathbf{1}$
$\gamma_\mu(V)$	$\gamma_\mu\gamma_5$	
$\gamma_\mu\gamma_5(A)$	γ_μ	
$\sigma_{\mu\nu}(T)$		$\sigma_{\mu\nu}\gamma_5$
$\sigma_{\mu\nu}\gamma_5(X)$		$\sigma_{\mu\nu}$

Γ	Reference Name	Abbr.	Symmetry Correspondences		
$\mathbb{I}$	Scalar	S	$U(1)_A$		
γ_5	Pseudo Scalar	PS			
γ_1,γ_2	Vector	V	$SU(2)_L \times SU(2)_R$		$SU(2)_{CS}$
$\gamma_1\gamma_5,\gamma_2\gamma_5$	Axial Vector	A			
$\gamma_4\gamma_3$	Tensor	T_t	$U(1)_A$		
$\gamma_4\gamma_3\gamma_5$	Axial Tensor	X_t			

Domain-Wall Fermions (DWF)

- **Localized Chiral States**: MDWF realizes 4D chiral fermions as boundary-localized states in the 5th dimension ($s \in [1, L_s]$):

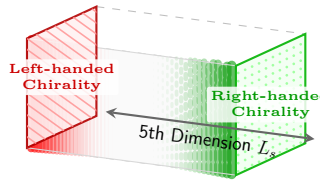
$$D_{\mathrm{DW}}^{4\mathrm{D}}(m) = \frac{1+m}{2} + \frac{1-m}{2}\gamma_5 \tanh\left(L_s \tanh^{-1}(H_M)\right)$$

- **Residual Mass and Chiral Breaking**: At finite L_s, residual overlap between boundaries induces m_{res}:

$$\overline{\psi}\Delta_L\psi \simeq m_{\mathrm{res}}\overline{\psi}\psi + O(A, \bar{\psi}, \psi) + \mathcal{O}(a^2)$$

$$\delta_{\mathrm{chiral}} S_{4\mathrm{D}} = i\alpha m_q \bar{\psi}(\gamma_5 + \hat{\gamma}_5)\psi + 2i\alpha\bar{\psi}\gamma_5\Delta_L\psi$$

$$\Delta_L = \frac{1}{4}(1-\epsilon_L^2)$$



$$D_{s,s'}^{5\mathrm{D}}(m) = D_+^{(s)}\delta_{s,s'} + D_-^{(s)}(P_+\delta_{s-1,s'} + P_-\delta_{s+1,s'}) - m(D_-^{(1)}P_+\delta_{s,1}\delta_{s',L_s} + D_-^{(L_s)}P_-\delta_{s,L_s}\delta_{s',1})$$

where $D_+^{(s)} = b_5 D_W + 1$, $D_-^{(s)} = c_5 D_W - 1$, $P_\pm = \frac{1\pm\gamma_5}{2}$ $(b_5 = 1.5, c_5 = 0.5)$.

Residual Mass & Renormalization Setup

We measure the residual mass am_{res} from the midpoint pseudo-scalar density J_{5q} and apply the scalar renormalization factor $Z_m(\beta)$:

$$am_{\text{res}} = \frac{\sum_{\vec{x}} \langle J_{5q}(\vec{x}, t) P(\vec{0}, 0) \rangle}{\sum_{\vec{x}} \langle P(\vec{x}, t) P(\vec{0}, 0) \rangle}$$

β	4.13	4.15	4.17	4.18	4.20	4.23	4.30	4.405
T (MeV)	138.23	145.75	153.31	157.03	164.55	176.43	202.52	241.60
am_{res}	7.31e-4	4.98e-4	3.40e-4	2.80e-4	1.91e-4	1.08e-4	2.81e-5	3.75e-6
m_{res}/m_l	0.9087	0.5360	0.3394	0.2744	0.1836	0.1041	0.0299	0.0049
m_{res} (MeV)	1.61	1.16	0.832	0.703	0.503	0.304	0.0908	0.0145
$Z_m(\beta)$	0.9377	0.9524	0.9662	0.9729	0.9857	1.0038	1.0422	1.0927

Lattice Setup & Ensembles

- **Physical Point Scan ($48^3 \times 16$)**: 8 couplings $\beta \in [4.13, 4.405]$ ($N_{\text{cfg}}^{\text{cond}} \approx 2000$–$4855$ for stochastic Z_2 condensate/susceptibility; $N_{\text{cfg}}^{\text{meson}} \approx 400$ with 16-source $\times$ directional averaging for correlators).
- **Fixed $\beta = 4.17$ Multi-Mass & Finite-N_t Scan**: $a^{-1} = 2453$ MeV ($am_s = 0.040$), varying $am_l \in \{0.0020, 0.0035, 0.0070, 0.0120\}$ and $N_t \in \{12, 14, 16, 18\}$ with larger spatial volumes ($40^3 \times 16, 48^3 \times 18$) near/below T_c.

Physical Point LCP Setup ($48^3 \times 16$, $N_f = 2 + 1$)

β	T (MeV)	am_s	am_l	$N_{\text{cfg}}^{\text{cond}}$	$N_{\text{cfg}}^{\text{meson}}$ (16-src)
4.13	138.23	0.043547	0.000805	2000	395 (6320)
4.15	145.75	0.040843	0.000930	2024	398 (6368)
4.17	153.31	0.038400	0.001001	2008	396 (6336)
4.18	157.03	0.037265	0.001022	2029	402 (6432)
4.20	164.55	0.035150	0.001041	2072	407 (6512)
4.23	176.43	0.032315	0.001033	2012	395 (6320)
4.30	202.52	0.026930	0.000939	2052	404 (6464)
4.405	241.60	0.021032	0.000760	4855	—

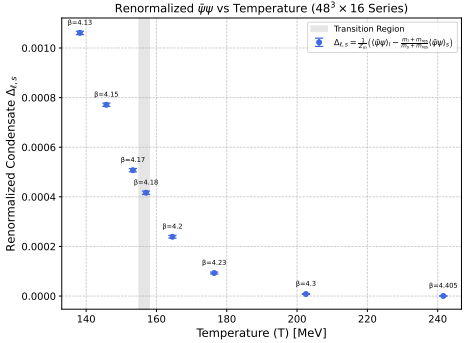
Fixed $\beta = 4.17$ Setup ($a^{-1} = 2453$ MeV)

Lattice Size	am_s	am_l	T (MeV)
$32^3 \times 12$	0.040	0.0020, 0.0035, 0.0070, 0.0120	204.00
$32^3 \times 14$	0.040	0.0020, 0.0035, 0.0070, 0.0120	174.86
$40^3 \times 16$	0.040	0.0020, 0.0035, 0.0070, 0.0120	153.00
$32^3 \times 16$	0.040	0.0020, 0.0035, 0.0070, 0.0120	153.00
$48^3 \times 18$	0.040	0.0020, 0.0035, 0.0070, 0.0120	136.00
$36^3 \times 18$	0.040	0.0020, 0.0035, 0.0070, 0.0120	136.00

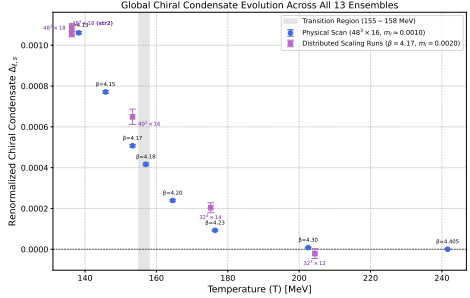
Renormalized Subtracted Chiral Condensate

- To eliminate additive UV divergences $\propto m_q/a^2$ and residual chiral breaking m_{res}, we construct the renormalized subtracted condensate evaluated with Z_2 stochastic volume sources:

$$\Delta^R_{\ell,s}(T) = \frac{1}{Z_m(\beta)}\left[\langle\bar{\psi}\psi\rangle_\ell - \frac{m_\ell + m_{\text{res}}}{m_s + m_{\text{res}}}\langle\bar{\psi}\psi\rangle_s\right], \qquad \langle\bar{\psi}\psi\rangle_{\text{phys}} = (N_t T)^3 \Delta^R_{\ell,s}(T)$$

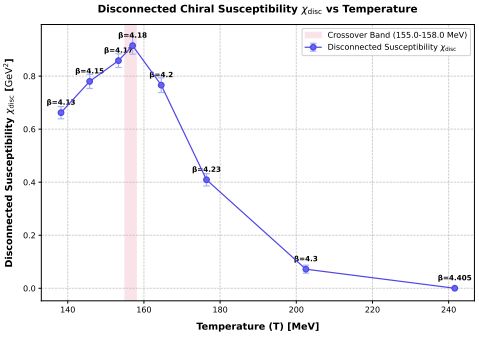


Renormalized condensate $\Delta^R_{\ell,s}$ on $48^3 \times 16$



Global comparison: $48^3 \times 16$ ($m_l \approx 0.001$) vs $\beta = 4.17$ ($m_l = 0.002$)

Disconnected Chiral Susceptibility & T_{pc}



χ_{disc} [GeV^2] vs T on $48^3 \times 16$ lattices

- **Unbiased Quadratic Estimator**: For k stochastic sources $O_{\ell,i} = \text{Tr}(D_\ell^{-1})_i$, diagonal noise bias is removed via:

$$\overline{O^2} = \frac{1}{k(k-1)} \sum_{i \neq j} O_{\ell,i} O_{\ell,j}$$

$$\chi_{\text{disc}} = (N_t T)^2 V \left(\langle \overline{O^2} \rangle - \langle \bar{O} \rangle^2 \right)$$

- **Sharp Crossover Peak**: Reaches global maximum at $\beta = 4.18$ ($T = 157.03$ MeV):

$$\chi_{\text{disc}}(157.0 \text{ MeV}) = 0.915(32) \text{ GeV}^2$$

$$\chi_{\text{disc}}/Z_m^2 = 0.967(34) \text{ GeV}^2$$

- **Padé** [3/2] **Rational Fit**: Yields $T_{\text{peak}} = 155.1(1.1)$ MeV ($\chi^2/\text{dof} = 0.69$), in excellent agreement with the HotQCD crossover band $T_{pc} = 156.5 \pm 1.5$ MeV.

Extracting Screening Masses: Methodology

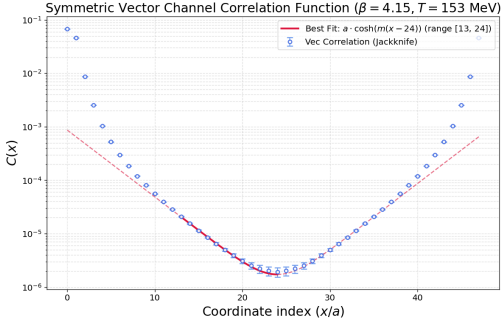
- **Multi-Source & Directional Averaging**: Correlators are averaged over 16 spatial sources and all equivalent polarization/spatial directions (x, y, z), preserving signed correlators prior to folding around $N_s/2$.
- **Local Effective Mass**: Solved numerically via vectorized Newton-Raphson at each separation x to identify a flat ground-state plateau window $[x_{\min}, x_{\max}]$:

$$\frac{C(x)}{C(x+1)} = \frac{\cosh(m_{\text{eff}}(x)(x - N_s/2))}{\cosh(m_{\text{eff}}(x)(x + 1 - N_s/2))}$$

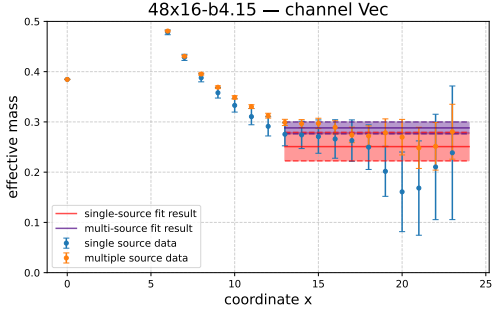
- **Screening Mass Fit & Jackknife**: m_{scr} is extracted by a least-squares χ^2 fit on each binned Jackknife sample to eliminate HMC autocorrelation bias:

$$C(x) = A \cosh\left(m_{\text{scr}}\left(x - \frac{N_s}{2}\right)\right)$$

Extracting Screening Masses: Fit Examples



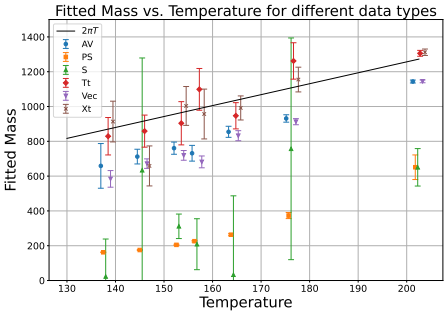
Symmetric Vector channel correlator for $\beta = 4.15$ $(48^3 \times 16)$



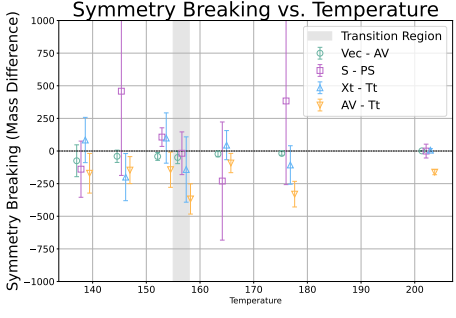
Effective mass plateau: 16-source vs single-source $(\beta = 4.15)$

Numerical Results at Physical Point ($48^3 \times 16$)

Grey band denotes the crossover region $T_{pc} = 156.5 \pm 1.5$ MeV:



Screening masses across 6 meson channels vs T

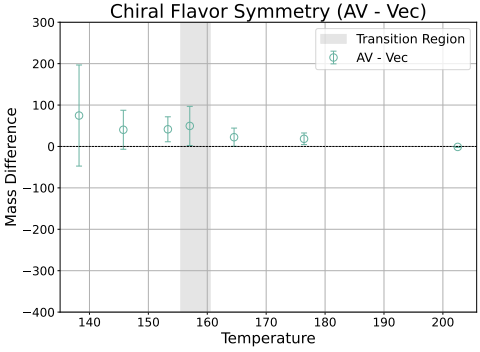


Symmetry breaking mass differences ΔM vs T

At $\beta = 4.30$ ($T = 202.5$ MeV): $aM_{\mathrm{Vec}} = 0.3533(10) \approx aM_{\mathrm{AV}} = 0.3529(12)$, $aM_{\mathrm{PS}} = 0.2009(25) \approx aM_{\mathrm{S}} = 0.2008(24)$, $aM_{\mathrm{Xt}} = 0.4049(38) \approx aM_{\mathrm{Tt}} = 0.4029(36)$.

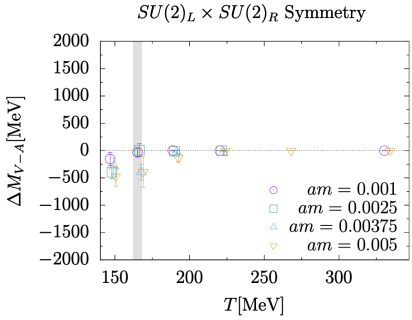
$SU(2)_L \times SU(2)_R$ Chiral Symmetry

Our Result ($N_f = 2 + 1$ MDWF)



Restoration behavior of chiral symmetry ($AV - Vec$)

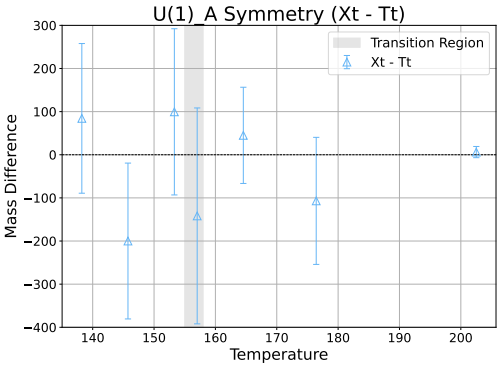
Previous Result ($N_f = 2$ MDWF)



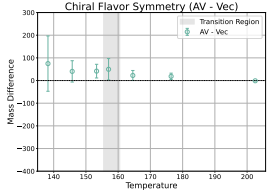
For comparison [PRD **111**, 114506 (2025)]

$U(1)_A$ Symmetry $(X_t - T_t)$

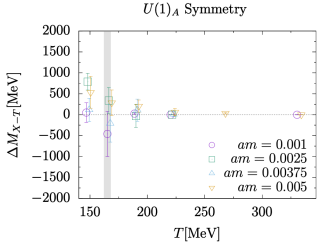
Our $U(1)_A$ ($N_f = 2 + 1$ MDWF)



Our Chiral ($N_f = 2 + 1$ MDWF)

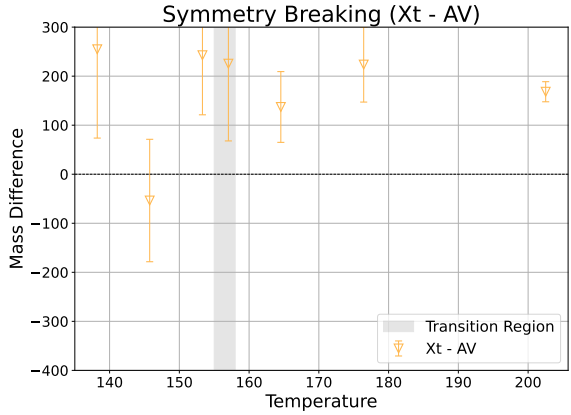


Previous $U(1)_A$ ($N_f = 2$ MDWF)



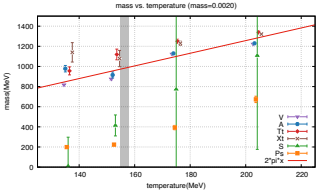
$SU(2)_{CS}$ Chiral-Spin Symmetry

- Tested via the mass splitting between X_t and AV (or AV and T_t) channels.
- Remains significantly broken up to $T = 202.5$ MeV ($\Delta M_{X_t - AV} \approx 164(19)$ MeV).
- Indicates that non-perturbative chromomagnetic interactions remain active in this temperature window.

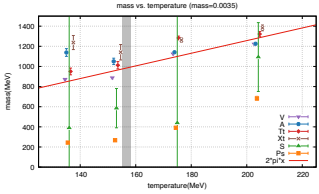


Screening Masses at Fixed $\beta = 4.17$

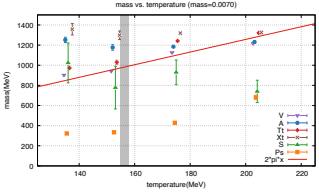
Multi-source screening masses vs $T \in [136.0, 204.0]$ MeV ($48^3 \times 18, 40^3 \times 16, 32^3 \times 14, 32^3 \times 12$, $a^{-1} = 2453$ MeV):



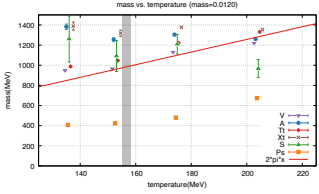
$am_l = 0.0020$



$am_l = 0.0035$



$am_l = 0.0070$

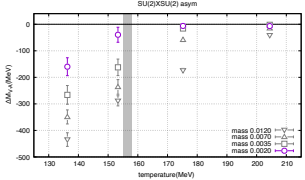


$am_l = 0.0120$

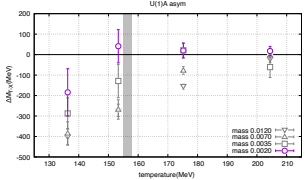
- Below T_c (136 MeV, $48^3 \times 18$), V–A and T_t–X_t split clearly while the pion (PS) remains light.
- Above T_c (174.9, 204.0 MeV), chiral partners (V–A) and axial partners (T_t–X_t) become degenerate across all four quark masses.
- High-T vector/tensor masses approach the free thermal asymptote $2\pi T$.

Symmetry Restoration at Fixed $\beta = 4.17$

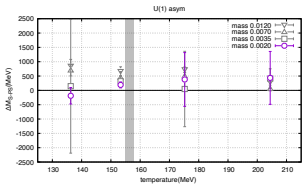
Mass differences ΔM vs $T \in [136.0, 204.0]$ MeV for $am_l \in \{0.0020, 0.0035, 0.0070, 0.0120\}$ ($am_l = 0.0020$ highlighted in purple):



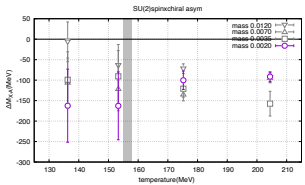
$SU(2)_L \times SU(2)_R$ (ΔM_{V-A})



$U(1)_A$ (ΔM_{T-X})



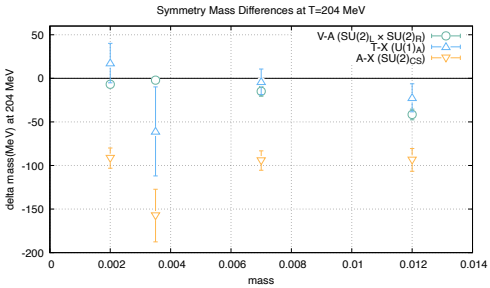
$U(1)_A$ (ΔM_{S-PS})



$SU(2)_{CS}$ (ΔM_{X-A})

- $SU(2)_L \times SU(2)_R$ $(V-A)$: Monotonically approaches zero above T_c; for $am_l = 0.0020$, ΔM_{V-A} drops from $-159.9(33.7)$ MeV (136 MeV) to $-6.8(4.0)$ MeV (204 MeV).
- $U(1)_A$ $(T-X)$: Simultaneously restores above T_c, reaching $+17.5(22.6)$ MeV at 204 MeV ($am_l = 0.0020$).
- $SU(2)_{CS}$ $(X-A)$: Remains clearly non-zero ($\approx -92(12)$ MeV at 204 MeV).

Chiral Limit Extrapolation ($T \approx 204$ MeV)



ΔM (MeV) vs am_l at $T = 204.0$ MeV ($32^3 \times 12$, $\beta = 4.17$)

- High-temperature snapshot at $T = 204.0$ MeV ($32^3 \times 12$) across $am_l \in \{0.0020, 0.0035, 0.0070, 0.0120\}$:

am_l	ΔM_{V-A} (MeV)	ΔM_{T-X} (MeV)	ΔM_{X-A} (MeV)
0.0020	-6.84 ± 3.96	$+17.48 \pm 22.64$	-91.56 ± 11.60
0.0035	-2.10 ± 1.34	-60.92 ± 51.07	-157.45 ± 30.17
0.0070	-14.91 ± 5.62	-3.98 ± 14.64	-94.35 ± 11.14
0.0120	-41.47 ± 6.12	-22.34 ± 16.08	-93.57 ± 13.04

- As $am_l \to 0$, both ΔM_{V-A} ($SU(2)_L \times SU(2)_R$) and ΔM_{T-X} ($U(1)_A$) extrapolate smoothly to zero.
- In contrast, ΔM_{X-A} ($SU(2)_{CS}$) remains around -92 MeV even in the chiral limit.

Summary and Outlook

- **Chiral Crossover & T_{pc}**: Renormalized subtracted chiral condensate $\Delta^R_{\ell,s}$ and disconnected chiral susceptibility $\chi_{\rm disc}$ on $48^3 \times 16$ MDWF lattices identify a smooth crossover with a peak at $\beta = 4.18$ ($T = 157.0$ MeV; Padé fit $T_{\rm peak} = 155.1(1.1)$ MeV), consistent with HotQCD (156.5 ± 1.5 MeV).
- **Physical Point Symmetry Restoration**: Above T_{pc}, $SU(2)_L \times SU(2)_R$ (V–AV) and axial $U(1)_A$ (T_t–X_t, S–PS) screening mass splittings vanish within uncertainties at $T = 202.5$ MeV.
- **Fixed $\beta = 4.17$ Chiral Limit Study**: Using 16-source signed correlators across $T \in [136, 204]$ MeV ($48^3 \times 18$ to $32^3 \times 12$), both ΔM_{V-A} and ΔM_{T-X} extrapolate to zero as $am_l \to 0$ at $T = 204$ MeV, supporting simultaneous restoration of $SU(2)_L \times SU(2)_R$ and $U(1)_A$.
- **Chiral-Spin Symmetry** $SU(2)_{CS}$: Remains broken up to 204 MeV ($\Delta M_{X-A} \approx -92(12)$ MeV), indicating active chromomagnetic interactions in the QGP regime just above T_c.
- **Outlook**: Improve statistics for the noisy scalar (S) channel near T_c and extend to finer lattice spacings toward the continuum limit.

Backup Slides

Backup: Overlap Dirac Operator

The massless operator $D_{\rm ov}(0) = \frac{1}{2}(1 + \gamma_5 {\rm sgn}(H_W))$ with Hermitian H_W satisfies the Ginsparg-Wilson relation:

$$\gamma_5 D_{\rm ov} + D_{\rm ov}\gamma_5 = 2 D_{\rm ov}\gamma_5 D_{\rm ov}$$

The massive overlap operator is defined as:

$$D_{\rm ov}(m) = D_{\rm ov}(0) + m(1 - D_{\rm ov}(0)) = \frac{1+m}{2} + \frac{1-m}{2}\gamma_5 {\rm sgn}(H_W)$$

$$\widehat{\gamma}_5 = \gamma_5 \left(1 - 2\not{D}_{\rm ov}(0)\right) = -\gamma_5 {\rm sgn}(H_W) \tag{1}$$

which is hermitian and satisfies $\widehat{\gamma}_5^2 = 1$ by virtue of the Ginsparg-Wilson relation. The lattice chiral transformation is defined as:

$$\delta\psi = i\alpha\widehat{\gamma}_5\psi, \quad \delta\overline{\psi} = i\alpha\overline{\psi}\gamma_5 \tag{2}$$

Backup: Order of Chiral Phase Transition & Scenarios

- Perturbative $4-\epsilon$ expansion RG analysis of effective Φ^4 linear sigma model shows no stable IR fixed point for $N_f > \sqrt{3}$ flavor symmetry [Pisarski-Wilczek (1984)].
- For $N_f \geq 3$, the anomaly determinant term doesn't change the order of the transition; for $N_f = 2$, it becomes a quadratic mass-like term which can smooth out the singularity and make the transition 2nd-order.
- **Previous Scenario**: 1st-order if $U(1)_A$ is restored at T_c; 2nd-order O(4) if $U(1)_A$ remains broken.
- **Recent Higher-Loop / FRG / Conformal Bootstrap Scenarios**: O(4) if $U(1)_A$ remains broken; weak 1st-order or 2nd-order ($\mathrm{O}(4) \times \mathbb{Z}_2$, $\mathrm{O}(4) \times \mathbb{Z}_4$, $\mathrm{O}(4) \times \mathrm{O}(2)$) if $U(1)_A$ is restored [Pelissetto-Vicari (2013), Nakayama-Ohtsuki (2016), Fejos (2022)].

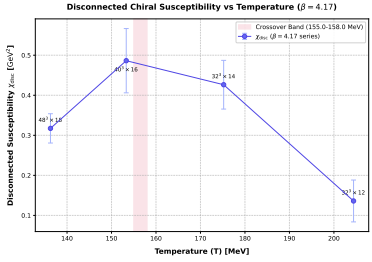
Backup: Condensate & Susceptibility Data Tables

Standard $48^3 \times 16$ LCP Scan

β	T [MeV]	$\Delta^R_{\ell,s}$	$\langle\bar{\psi}\psi\rangle^R_{\text{sub}}$ [GeV^3]	χ_{disc} [GeV^2]
4.13	138.23	$1.0608(67) \times 10^{-3}$	$1.148(7) \times 10^{-2}$	0.662(23)
4.15	145.75	$7.7109(68) \times 10^{-4}$	$9.779(86) \times 10^{-3}$	0.780(27)
4.17	153.31	$5.0728(66) \times 10^{-4}$	$7.487(98) \times 10^{-3}$	0.858(26)
4.18	157.03	$4.1649(66) \times 10^{-4}$	$6.606(105) \times 10^{-3}$	0.915(32)
4.20	164.55	$2.3896(56) \times 10^{-4}$	$4.361(102) \times 10^{-3}$	0.766(28)
4.23	176.43	$9.2886(381) \times 10^{-5}$	$2.089(86) \times 10^{-3}$	0.409(23)
4.30	202.52	$8.204(1330) \times 10^{-6}$	$2.791(451) \times 10^{-4}$	0.072(15)
4.405	241.60	$4.030(207) \times 10^{-7}$	$2.328(120) \times 10^{-5}$	$8.6(1.7) \times 10^{-5}$

Fixed $\beta = 4.17$ Scaling ($am_l = 0.0020$)

Size	T [MeV]	$\Delta^R_{\ell,s}$	χ_{disc} [GeV^2]
$48^3 \times 18$	136.28	$1.055(14) \times 10^{-3}$	0.317(37)
$40^3 \times 16$	153.31	$6.491(38) \times 10^{-4}$	0.486(80)
$32^3 \times 14$	175.21	$2.039(24) \times 10^{-4}$	0.426(61)
$32^3 \times 12$	204.41	$-2.10(242) \times 10^{-5}$	0.136(52)



Backup: Effective Mass Plateaus ($48^3 \times 18$, $\beta = 4.17$)

